

American Put Pricing Model Report

Github Repository Link

<https://github.com/workkrishnpatel/IIT-B-ITSP-SOC-2026>

Story till now:

Till Week 5 we have seen the theory and some practical aspects of the project the function we have defined and did in week 4 will be used in Week 6 during model training , the binomial crr pricer function we worked on that is the essential function for the labeling the model this week 6 .

Overview of Week 6

in week 6 we first created the synthetic dataset for our American puts option pricing then we have used the neural nets for the modeling the pricing using the MLP i.e. Multi Layer Persceptrone . wthe details of the model is mentioned in the following sections .

Dataset Information

for week 6 , american puts pricing model we have created the synthetic data in the particular range the range is as follow :

Parameter Ranges

S_0 (Initial Stock Price): 60 to 140

K (Strike Price): 80 to 120

T (Time to Maturity): 0.05 to 2.00 years

r (Risk-Free Interest Rate): 0.00 to 0.10 (0% to 10% per annum)

σ (Volatility): 0.10 to 0.50 (10% to 50% annualized)

in the range above we have created the dataset using the uniform distribution using numpy and later all numpy arrays converted into dataset the dataset is available at the project github folder [Week_6](#)

<https://github.com/workkrishnpatel/IIT-B-ITSP-SOC-2026> . after the dataset is created we have used the american crr binomial function we have used in the Week 4 assignment . so for each array or row of the dataframe we applied the function for the american put price and for all the rows we have created the price and since there are 25,000 rows it will take some time for getting fetched the Puts_Pricer . and finally our dataset is ready with the shape of 25000,6

Sanity Check for the dataset before Modeling

before modeling with this dataset we need to check some sanity for avoiding data leakage and also to avoid wrong data input . After dataset is formed we have checked the following sanity check

1. the puts price must be greater than 0
2. there should not be any value which is infinity
3. it should not exceed the intrinsic value

so these are the sanity checks we need to perform before modeling , there must not be any error in the parameter ranges since in the function of the crr pricer function we have handled the errors for the parameters which are not in proper format and all .

Modeling

after all the sanity check , we are good to go for modeling with dataset .since this parameters are non linear we can't use the single perceptrone , instead of that we need to use MLP or ANN which is better for capturing non linear inferences from dataset so we are using the tensorflow framework of deep learning and hence keras library . we have added 4 hidden layers in our model and 1 input and output layer . since our model is regression analytic linear activation function is better in the output layer and elsewhere ReLU . ReLU IS CONSIDER AS THE BEST ACTIVATION FUNCTION OF ALL TIME SINCE IT IS OUTSTANDS IN COMPARISON OF OTHERS . NEVERTHELESS , WE HAVE ADDED 128 , 64,32,16 NODES IN THE LAYERS INCREASING SO THEN WE TRAINED THE MODEL WITH ADAM OPTIMIZER WHICH IS AGAIN CONSIDER AS THE BEST OPTIMIZER TILL DATE .

THE NOTEBOOK FOR THE DATASET GENERATION AND ALSO FOR THIS MODELING IS AVAILIABLE IN MY GITHUB REPOSITORY <https://github.com/workkrishnpatel/IIT-B-ITSP-SOC-2026>.

Results

before we will get on the results , lets discuss the extra configuration during the modeling . i have used 200 epoches with batch size of 32 and also used calls backs like early stopping and decreasing learning rate upon plato curve .

MAE : 0.031422

RMSE : 0.044302

R^2 : 0.999989

Max Absolute Error : 0.434644

this are amazing results we actually got , the four parameters metrics are excellent , the first one MAE i.e. Mean Absolute error is the abeage of all absolute error . the best balue of MAE is 0 which indicate no error between prediction and actual .so since our MAE is almost 0 hich implies model is performing well and doing bery

less mistakes , and the RMSE is nothing but the root of MAE so signifies the same . the max absolute error is 0.4 approx in the some balue set of data we got the max error which is also significantly less but this si the max error we got and the top 10 max absolute error is mentioned below . the r2 score is rhe accuracy metric is meant for regression problem settings and as you can see that it is almost perfect and that implies the model is performing very well .

always rememver the r2 score doesn't implies that your model is good we should also look upon test data . i am sure that my model is good because my model is neighter over fits nor under fits . the graphs of the data is mentioned or attached below .

Top 10 maximum absolute error datapoints and its values

Rank	Sample Index	Actual Price	Predicted Price	Absolute Error
1	5732	4.104398	4.539042	0.434644
2	5001	4.745776	5.035478	0.289701
3	9655	13.751834	13.464457	0.287377
4	16403	12.337801	12.071216	0.266586
5	11056	4.376589	4.640084	0.263495
6	5068	19.266735	19.060907	0.205828
7	15318	4.053020	3.847204	0.205816
8	23797	3.750121	3.939250	0.189130
9	752	14.311544	14.126458	0.185086
10	8604	7.243555	7.424785	0.181230

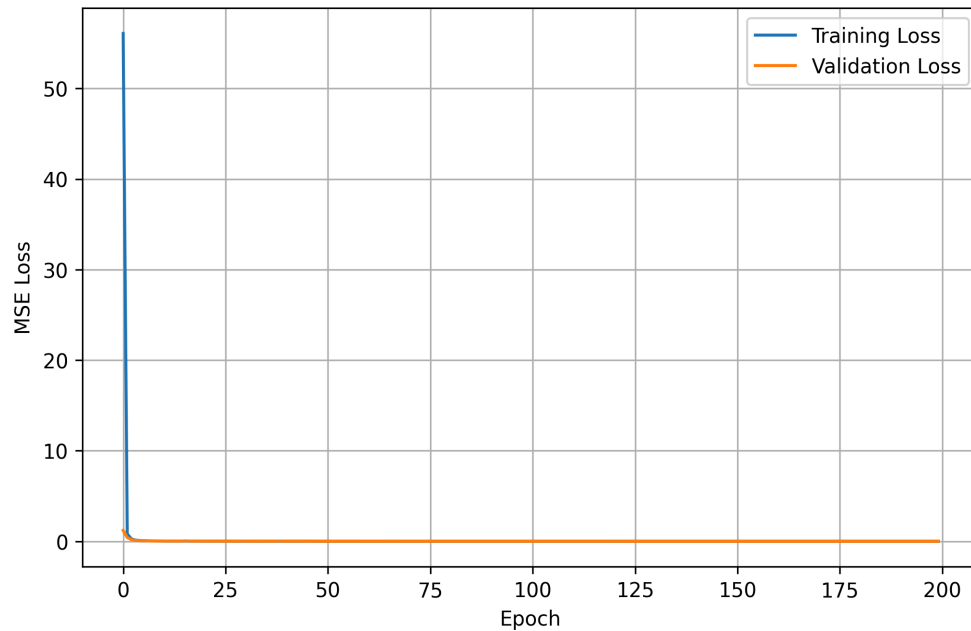
you can take look at the dataset points by using the following line of code

python

```
df.iloc[index]
```

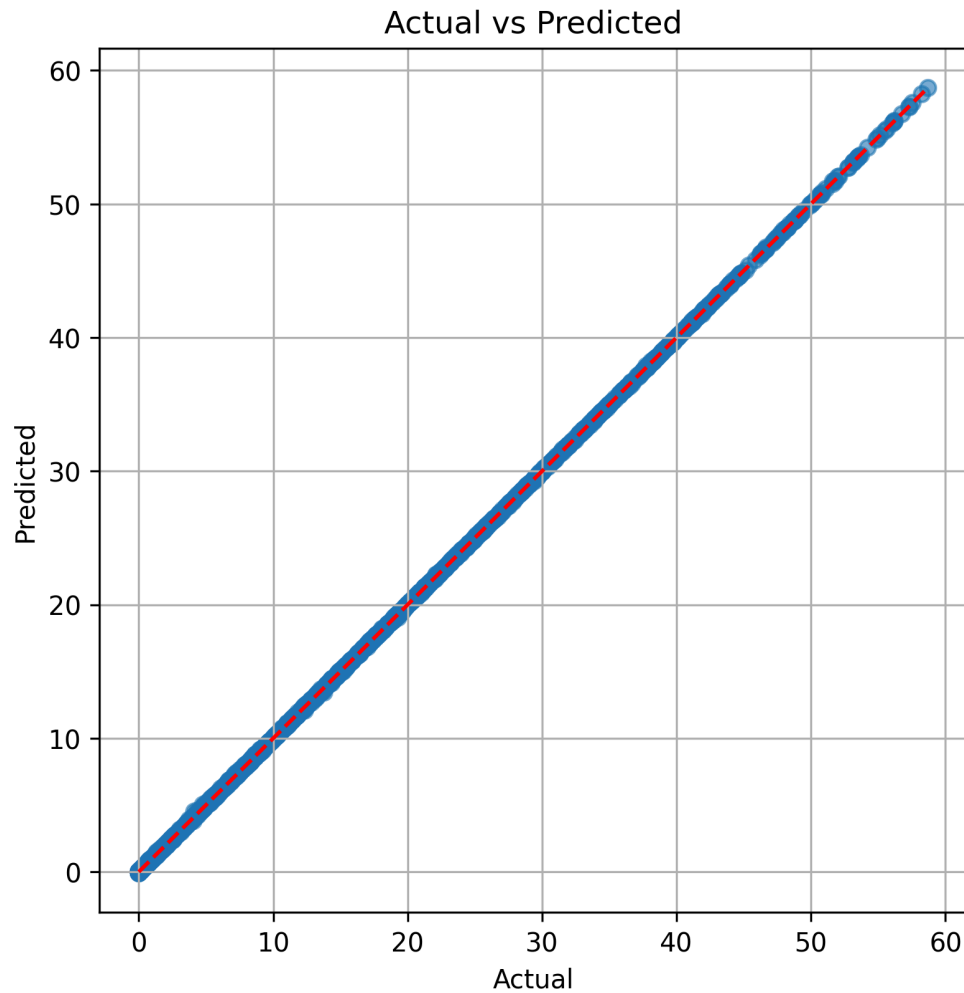
Lets look and dive into the graphs of the model and interpret it .

Graph 1 : Training Curve vs. Validation Loss Curve



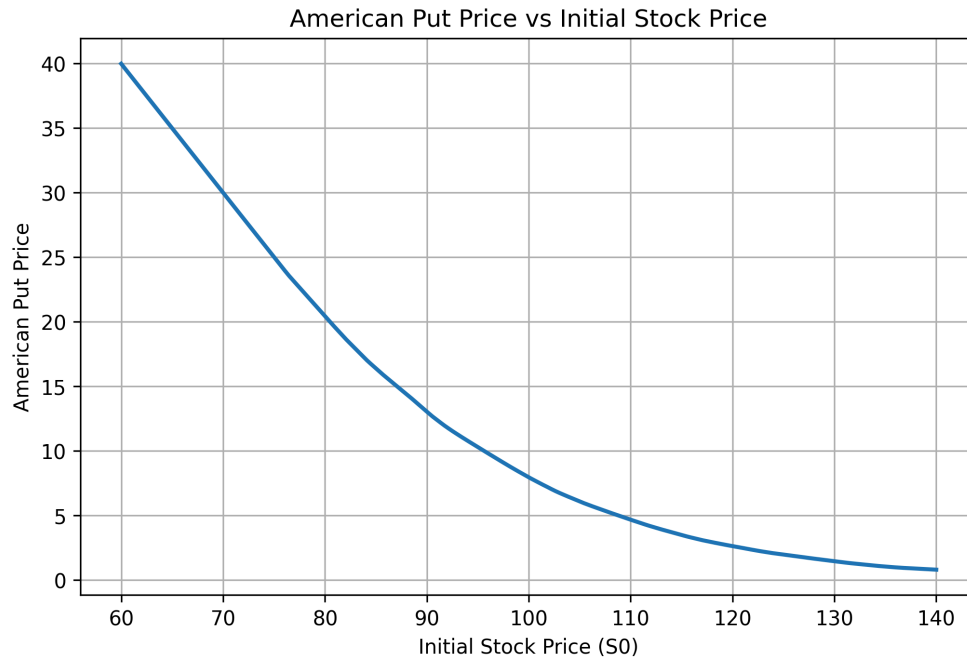
as you can see till epochs 50 the model significantly reduces the loss and in very fast manner and then after it showed up stiff and plato which implies the model is learning well ,especially the validation loss so thats a good sign .

Graph 2: Prediction vs, Actual Scatter Plot



as you can see this scatter plot this scatter plot points are going from left bottom to right top which signifies positive correlation between the actual and prediction and almost linear relationship between them .

Graph 3: Spot Price vs. Put Price



as mentioned in the sanity check , this is the important sanity check that whenever the spot price increases the puts price decreases , hence it is clearly visible in the graph that the same . which concludes the succession of the sanity check

Conclusion

The neural network demonstrated exceptional performance in approximating the American Put option prices generated by the CRR Binomial Tree model. On the unseen test dataset, the model achieved a Mean Absolute Error (MAE) of **0.0314**, a Root Mean Squared Error (RMSE) of **0.0443**, and an **R² score of 0.999989**, indicating an almost perfect fit between the predicted and actual option prices. The maximum absolute prediction error observed across the entire test set was only **0.4346**, demonstrating that even the worst-case prediction deviated by less than half a monetary unit.

The learning curves showed stable convergence of both training and validation losses without signs of overfitting, while the Actual vs. Predicted plot closely followed the ideal 45-degree reference line, confirming the model's excellent

predictive capability. Furthermore, the residual distribution was centered around zero, indicating unbiased predictions with minimal systematic error. The financial sanity check also verified that the predicted American put prices decrease monotonically as the underlying stock price increases, confirming that the trained model preserves the expected financial behavior of American put options.

Overall, the developed Keras-based neural network serves as an accurate and computationally efficient surrogate for the CRR Binomial Tree model, providing near-instantaneous option price predictions while maintaining outstanding predictive accuracy. These results demonstrate the effectiveness of deep learning as an alternative to traditional numerical option pricing methods for real-time financial applications.

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